A SQUARE-ZERO UNITIZATION WITH NON-OPEN MULTIPLICATION

AGENT LANE 05

Source And Target

The source paper is Draga–Kania, *When is multiplication in a Banach algebra open?*, arXiv:1704.08608 [1]. Open Problem 3 on page 14 asks:

> Do commutative, unital Banach algebras with zero-dimensional maximal ideal space have (uniformly) open multiplication? What if idempotents in the algebra are linearly dense?

(3) Do commutative, unital Banach algebras with zero-dimensional maximal ideal space have (uniformly) open multiplication? What if idempotents in the algebra are linearly dense?

Result

The first question has a negative answer as written. There is a commutative unital complex Banach algebra whose maximal ideal space is a singleton, but whose multiplication is not open. Consequently its multiplication is not uniformly open.

The example is non-semisimple and has only the trivial idempotents 0 and 1. Thus it does not answer the stronger follow-up question about linearly dense idempotents.

Proof Intuition

Take the unitization of a square-zero two-dimensional ideal. Since every element of the ideal is nilpotent, every character vanishes on that ideal, so the maximal ideal space is just one point. Openness fails for a geometric reason. Near a fixed nilpotent vector e_1, both factors have ideal parts close to e_1. To produce a product with scalar coordinate 0, one of the scalar coordinates must be exactly 0. The ideal part of the product then becomes a scalar multiple of a vector still close to e_1, and so it cannot equal a small nonzero multiple of a transverse vector e_2.

Theorem 1. *There exists a commutative unital complex Banach algebra A such that $\mathrm{Max}(A)$ is zero-dimensional and multiplication*

$$A \times A \longrightarrow A, \qquad (a, b) \mapsto ab,$$

is not open.

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Proof. Let $E = \mathbb{C}^2$, equipped with the norm $\|(z_1, z_2)\|_\infty = \max\{|z_1|, |z_2|\}$. Let

$$A = \mathbb{C} \oplus E$$

with norm $\|(\lambda, x)\| = |\lambda| + \|x\|_\infty$ and multiplication

$$(\lambda, x)(\mu, y) = (\lambda\mu, \lambda y + \mu x).$$

This is the usual unitization of the square-zero algebra E, where $xy = 0$ for all $x, y \in E$. The displayed norm is submultiplicative:

$$\begin{aligned}\|(\lambda, x)(\mu, y)\| &= |\lambda\mu| + \|\lambda y + \mu x\|_\infty \\ &\leq |\lambda||\mu| + |\lambda|\,\|y\|_\infty + |\mu|\,\|x\|_\infty \leq \|(\lambda, x)\|\,\|(\mu, y)\|\,.\end{aligned}$$

The algebra is plainly commutative and unital, with identity $(1, 0)$.

We first identify the maximal ideal space. If $\varphi : A \to \mathbb{C}$ is a character and $x \in E$, then $(0, x)^2 = 0$, hence

$$\varphi(0, x)^2 = \varphi((0, x)^2) = 0.$$

Thus $\varphi(0, x) = 0$ for every $x \in E$, and since $\varphi(1, 0) = 1$,

$$\varphi(\lambda, x) = \lambda.$$

So A has exactly one character. Its maximal ideal space is therefore a singleton, hence zero-dimensional.

It remains to prove that multiplication is not open. Let $e_1 = (1, 0)$ and $e_2 = (0, 1)$ in E, and put

$$a = (0, e_1) \in A.$$

We show that multiplication is not open at (a, a). Fix $0 < \varepsilon < 1$. We claim that no ball around $0 = a^2$ is contained in

$$B_A(a, \varepsilon) \cdot B_A(a, \varepsilon).$$

Indeed, let $\delta > 0$ and choose $0 < t < \delta$. Consider

$$z = (0, te_2) \in A.$$

Then $\|z\| = t < \delta$. Suppose, towards a contradiction, that

$$z = (\alpha, x)(\beta, y)$$

with $(\alpha, x), (\beta, y) \in B_A(a, \varepsilon)$. Comparing scalar coordinates gives

$$\alpha\beta = 0.$$

Hence $\alpha = 0$ or $\beta = 0$. If $\alpha = 0$, then the ideal coordinate of the product is $\beta x = te_2$. Since $t \neq 0$, we have $\beta \neq 0$, and therefore $x \in \mathbb{C}e_2$. But every vector in $\mathbb{C}e_2$ has distance at least 1 from e_1 in the supremum norm:

$$\|x - e_1\|_\infty \geq 1.$$

This contradicts

$$\|(\alpha, x) - a\| = |\alpha| + \|x - e_1\|_\infty < \varepsilon < 1.$$

The case $\beta = 0$ is identical, using y instead of x.

Thus every ball around 0 contains points not belonging to $B_A(a,\varepsilon)\cdot B_A(a,\varepsilon)$. Multiplication is not open at (a,a), and hence is not an open map. □

Verification Notes

No computational verification is involved. The key algebraic checks are:

- $0 \oplus E$ is a square-zero ideal, so all characters kill it;
- $\mathrm{Max}(A)$ is the singleton scalar character;
- the failure of openness is witnessed by the missing transverse points $(0, te_2)$ in every neighborhood of 0.

Novelty And Search Bounds

Before promotion, the run's registry, solution, attempt, and proof-gap indexes were searched for arXiv:1704.08608, the title, "zero-dimensional maximal ideal space", "square-zero", "unitization", and "open multiplication". No existing packet for this target was found.

External web searches on 2026-06-15 used exact and near-exact phrases including "zero-dimensional maximal ideal space open multiplication Banach algebra", "square-zero open multiplication Banach algebra", "unitization zero-dimensional maximal ideal space multiplication open", and the exact printed question from Open Problem 3. These searches did not find a later explicit answer. Novelty confidence is medium because the example is elementary and may be known folklore.

Human Review Recommendation

Review as a candidate counterexample to the first clause of Open Problem 3 as printed. The main semantic issue is whether the authors intended an unstated semisimplicity assumption. The dense-idempotent follow-up remains open in this packet.

References

[1] S. Draga and T. Kania, *When is multiplication in a Banach algebra open?*, arXiv:1704.08608, 2017.